

Community Detection and Centrality Measures in Alzheimer's Disease Network Analysis

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Abstract—This project intricately explores network analysis in the context of Alzheimer's disease, emphasizing community detection and centrality as a pivotal facet in data research on complex networks. Community structure, an essential component of complex networks, has progressively emerged as a focus of study interest. The community detection algorithm is useful for examining the network's true structure. The paper highlights the importance of community identification strategies in understanding network behavior and topology analysis by classifying them into non-overlapping and overlapping methods. Many ways are looked at, such as those built on different factors separation into parts, group linking, tag movement and gradual steps partitioning. The process of finding out what sicknesses, like Alzheimer's disease someone has also highlights the advantages of knowing and working with many different abilities, skills working as a group to fix issues, and sharing what we know in a community. It present a way to look at community splits using a measure called modularity. This focuses on how it can be used to check layout, steer refinement, and comprehend network dynamics. In this paper, we look into measures of importance, especially coreness centrality. We also talk about core-periphery structures, k-shell decomposition, and extensions like the all-around score and hybrid centrality measures, which provide useful tools for determining influential nodes in complex networks.

Index Terms—Social Network Analysis, Coreness Centrality, All Around Score, Synthesize Centrality, C - Index, Girvan Newman, Kernighan-Lin Bisection Algorithm, Louvain Algorithm and Spectral Algorithm.

I. INTRODUCTION

Alzheimer's disease is a progressive neurological illness that poses a significant challenge to scientific research and medical care. This condition, named for Dr. Alois Alzheimer, whose unique brain abnormalities were initially discovered in 1906, mainly affects cognitive functioning, memory, and activities of daily life. Alzheimer's [27] is a type of dementia that progressively deteriorates cognitive function. It affects people individually and also has a significant negative influence on carers and global healthcare systems.

The situation is marked by the collection of strange protein build-ups, mainly twisted strings (tau tangles) and clumps

made from beta-amyloid in our brain. These harmful changes make brain cells slowly break down, messing up how they talk to each other and leading to a loss in thinking ability. The sickness can move through different steps, going from small memory loss to big problems with thinking.

Alzheimer's sickness is becoming more and more important, as populations get older its numbers keep going up. As nothing can cure Alzheimer's right now, most of the study on this sickness is about finding out why it happens and making good ways to treat it. A main aim is understanding the complex mix of genes, surroundings and actions that add to Alzheimer's disease. Finding problems early, making sure what's wrong is clearly understood, and coming up with spot-on care methods are some of the challenges.

In the last few years, network analysis has been a helpful method for understanding complicated links in living things like our brain. We can learn a lot about Alzheimer's disease on a bigger scale by using network study in the area.

A. Dataset

The scientific community has open access to brain MRI data sets to introduce various Centrality and community detection and demonstrate its performance through the implementation of the proposed produce. This dataset contains the following attributes: Estimated Total Intracranial Volume, Normalise Whole Brain Volume, Mini Mental State Examination, Clinical Dementia Rating, Identification, Gender, Dominant Hand, Age in Years, Education Level, Socioeconomic Status.

A cross-sectional study of MRI data in middle-aged, non-demented, and demented older adults. Contains a variety of properties. This dataset has 416 cross-sectional participants, ranging in age from 18 to 96. For every patient, three or four distinct T1-weighted MRI scans from a single scan session are displayed. Among the subjects are both men and women, all of them are right-handed. In 100 of the individuals over 60, very mild to moderate Alzheimer's disease (AD) has been clinically diagnosed. A reliability data set with 20 nondemented participants photographed on a follow-up visit within 90 days after

the first session is also included. Each data point is associated with a relevant category based on pre-existing classifications, manual annotations, or prior knowledge.

B. Organization of Paper

The following describes the way the paper is organized: A review of the relevant research that came in to the light are included in section II, outlining the subject of the work. The methodology is explained in section III, where how does centrality and community detection has accomplished. The experimental results in which the results that we had done on the dataset and analysis is demonstrated in section IV, at last our study is ultimately concluded in section V, and section VI, future work is included the extension of the work that we what to archive.

II. RELATED WORK

The research explores intellectual endurance in folks with Alzheimer's disorder abnormality study, coined AsymAD, using a web-centric method [28]. Proteomics study pinpoints Neural Regeneration Protein 1 as a central node polypeptide linked with neural biotechnology and intellectual toughness. Trails in a cell-based framework of AD illustrate Neural Regeneration Protein 1's capability to advocate arborized vertebra strength and ease beta-amyloid-induced nerve cell overexcitability [28]. Additional examination uncovers coinciding synaptic bioscience connecting Neural Regeneration Protein 1 - driven transformations in nurtured nerve cells with routes correlated with cerebral robustness in individuals. Combining homo sapiens encephalon protein complement records with simulation platforms innovations awareness of fortitude systems, enabling in the determination of remedial destinations for neurodegenerative disease [28].

Senile dementia of Alzheimer's demonstrates a hard interplay of heredity and ecological factors. This research deploys an innovative strategy incorporating numerous information sets (GWL, GWA, GWE) to determine 108 reasonable Alzheimer's disease (AD) genetic information [29]. Peptide - peptide interaction web simulation spotlights substantial aggregates, with Epidermal Growth Factor Receptor and Actin, Beta (ACTB) burgeoning as pivotal hub nodes [29]. Operational elucidation discloses tasks in brain cell generation and Mitogen - activated protein kinase modulation of activity, while convergence with Cerebrospinal Fluid (CSF) and blood plasma protein profile discerns prospective molecular makers, accentuating the efficacy of this unified genetic integration method for listing by importance Alzheimer's disease contender hereditary factors [29].

AD is notable by beta-amyloid plaque and tau tangles, entailing peptides like Amyloid - beta 1 - 42 and tau hyperphosphorylation [30]. Bioorthogonal chemistry covalent linking and computational modeling of neural networks make known neurodegenerative disease-specific amino acid chain associations and prospective objectives for interrupting fibrillogenesis. Gene function classification systematically reviews

important findings powerhouse of cell and inner cell space contribution in AD-associated accumulation [30].

III. METHODOLOGY

To start with, we developed a network from the data we gathered which is the MRI dataset. Nodes in our network depict each single participant, while edges describe the mutual relations that are approved from the dataset characteristics. For example, edges show the common between the clinical dementia ratings or other related attributes. Our methodology consists of a wide range of centrality measures in order to identify the more involving and connecting nodes as well as performing community detection algorithms to uncover underlying network structures.

A. Centrality Measures

In 1948, Bavelas came up with the idea of centrality in how people communicate to measure who has more power (Bavelas, 1948) [1]. Consequently, there are four fundamental ideas related to centrality: the own line, in-between position, nearness and key importance [2]. Each way of measuring importance shows different things. When we judge an actor's importance based on how closely they link to other people or things, it is best to use a measure called degree centrality. When the power of a node is linked to its middleman role, we should use betweenness centrality.

Centrality, however, ought to be applied in situations where an actor's strength is correlated with their accessibility to other nodes [3]. In diffusion applications, eigenvector centrality should also be taken into account. Frequently, the subject is reduced and restricted to unweighted and undirected [4] networks. Nonetheless, there is no agreed-upon definition of an actor's centrality, not even for these rather straightforward graphs. Other centrality indicators have been proposed over time in addition to the fundamental ones listed above. Directed or weighted networks can be taken into consideration with these recently proposed centrality measures, some of which are designed for specific uses.

Relationships and trends within the dataset can be understood by performing a centrality analysis on a network created from the records of individuals who had their Alzheimer's illness checked. In this case, the network's nodes may stand in for specific people, and the edges could indicate different connections or interactions between them. By examining centrality in a network of people with and without Alzheimer's, researchers can pinpoint particular people whose patterns of centrality are different in the two groups. This may result in the identification of early indicators or risk factors linked to the illness.

1) Coreness Centrality: The popular coreness centrality measure ignores the impact of topological links on nodes' spreading capacity and ranks nodes influence only by adding the k-shell values of all of their neighbors. Using the coreness of its neighbors, the coreness centrality measures a node's capacity for spreading. In this method, the network's center will be derived by the node which has more connections with

the other nodes or participants. Neighbors k-shell indices may serve as useful predictors of a node's potential for spreading. We can use the susceptible-infected recovered(SIR) method to inspect the spreading in order to evaluate the effectiveness of the suggested measure.

According to Kitsak et al., there are tenable situations in which the highest-betweenness or highly linked nodes have minimal bearing on the propagation of a particular spreading process. Rather, they demonstrated that the spreaders that are most effective are those that are found in the network's core and can be found by applying the k-shell [5] (also known as k-core) decomposition approach. Additionally, Zeng et al. pointed out that the linkages that connect to the exhausted nodes are completely ignored by the k-shell technique, [6] which only takes into account the links between the residual nodes. In order to take into account both the exhausted degree and the residual degree, they introduced the mixed degree decomposition (MDD) method. Liu et al. used an improved ranking technique to provide a more readable ranking list in terms of the node distance to the network core [7] (defined as the node set with the greatest k-shell values).

Formula for K-Core Centrality

$$C_k(n) = k$$

where:

- $C_k(n)$ - k-core centrality of node n
- n is the k-core level of node n

This approach, which assesses the rank correlations using Kendall's tau, is noticeably more accurate than previous centrality measurements.

2) **All-around Score:** For information to travel more effectively across complicated networks, identifying the most influential nodes is crucial [8]. The idea of an all-around score was first presented by Hou et al. to identify influential nodes.

$$d = \sqrt{|d|^2 + |C_B|^2 + |C_{sk}|^2}$$

where C_{sk} is the node's k-shell index, C_B is its betweenness centrality, and d is its degree. In order to determine a node's relevance, we take into account three metrics. The degree controls the node's local connection, betweenness handles the shortest paths that reflect global data, and k-shell indicates the node's location with relation to the center [9]. Because it depends on the betweenness centrality that has the maximum difficulty, the entire procedure has an $O(nm)$ time complexity. The overall distance may be a more reliable and accurate measure of a node's influence, according to the results.

3) **Synthesize Centrality:** Opinion leaders have an important influence on information propagation [10], so it is important to efficiently identify them to understand this dynamic phenomenon. Formula:

$$Scy = \frac{C_D + C_B}{C_C}$$

According to the experimental results, there is a strong likelihood that a node will be recognised as an opinion leader

using HITS and PageRank if it is recognised as one using SC centrality. The suggested approach is quite effective [11], and as the number of opinion leaders rises, so does its accuracy.

4) **C-Index:** In light of their investigation of researchers' proficiency in a weighted cooperation network, Yan et al. developed the C-index centrality metric. The degree, strength, and centrality data of the node's neighbors are used to calculate the c-index [12]. They demonstrate a strong correlation between the C-index and the cooperation network's competence. In weighted scale-free networks [13], it is distributed according to a power-law distribution. Additionally, they suggest the iterative c-index and cg-index as two additional extensions of the c-index centrality metric.

B. Community Detection

Community identification in complicated networks has become a hot topic. Consequently, the general public's interest in community organization has grown. When looking for natural subgroups or communities within a population, community detection for Alzheimer's disease can be helpful since it looks for similar traits or behaviors. Understanding social structures, recognizing at risk populations, developing customized interventions, and improving general insights into the various patterns and linkages found in the community under study are all made easier by this approach.

Nodes inside a community are more tightly related to one another than they are to nodes outside of it, based on Newman's 2002 concept of community structure. Whether or not community detection approaches overlap or do not overlap determines how they are classified. Each node inside a community has its own identity, and numerous separate communities can be identified. Compared to the not coinciding the community detection algorithm, the overlapping community discovery algorithm is more practically significant. The communities are initially connected to each other since overlapping nodes are the important nodes in the network. Second, overlapping communities are better able to represent the real-world network structure.

Quality checking is essential to community detection since it guarantees the authenticity, relevancy, and quality of the communities that are found. It offers a framework for analyzing network dynamics, improving community topologies, and assessing community identification techniques. To extract valuable insights from complicated networks and make decisions based on the communities found, quality checking is a crucial step.

The non-overlapping community structure is essentially the application domain for the early community detection techniques [16]. The overlapping community structure is the direction that the research field of community detection algorithms is taking with the introduction of the clique percolation technique [17]. This algorithm's basic tenet is to view the community as a collection of cliques. Any two nodes can form a clique, which is a complete subgraph where all cliques are connected by a common node.

In a complicated network, information is typically propagated between individuals by edges between nodes. It is recognized that nodes within a community share the same information [18], whereas nodes within different communities share different information based on community features. Thus, label propagation-based community detection techniques were developed. It is a semi-supervised learning technique based on graphs [19]. The algorithm comes to an end when the nodes' labels in the network stop changing. Lastly, nodes that share a label are members of the same community.

1) **Label Propagation:** The primary benefit of the label propagation method is its great execution efficiency due to its linear time complexity and lack of parameter requirements [20]. However, the system is limited to identifying community structures that do not overlap.

$$b_t(c, x) = \frac{b_{t-1}(c, y)}{|N(x)|}$$

When a node x iterates t times in a community c , its subordinate coefficient is represented by $b(c, x)$; its adjacent node set is represented by $N(x)$. Through testing, we are aware that the algorithm will continue to choose at random even in the presence of several labels. It causes the mining community's conditions to be unpredictable and the algorithm outcomes to be unstable. The method simultaneously restricts the number of communities that overlapping nodes can be a part of. The Wu et al. BMLPA expands upon COPRA by eliminating the cap on the number of communities overlapping [21] nodes can be a part of, while also enhancing the original algorithm.

2) **The Algorithms Based on Hierarchical Partitioning:** There are two types of hierarchical partitioning-based community discovery algorithms: split hierarchical technique and condensed hierarchical method. In 2004, Newman and Girvan [14] introduced the idea of modularity, allowing scholars to explore the concept of community quantitatively. Subsequent research employed modularity [15] as a metric to assess community partition quality. Better community structures connect with larger values. Girvan and Newman continually removed the edge with the maximum edge betweenness to develop the GN method [22], which is the structure of a hierarchical community.

a) **Girvan Newman:** To locate clusters or communities inside a network, the Girvan-Newman method is used as a community detection technique [23].

Important Girvan-Newman algorithm steps:

- Calculate Betweenness Centrality: Determine each edge in the network's betweenness centrality $C_k(v)$.
- Edge Removal: Eliminate the edge with the highest centrality betweenness. Iteratively removing high-betweenness edges causes the network's connectivity between various nodes to break.
- Recalculate Betweenness Centrality: Recalculate the betweenness centrality for the network's remaining edges following the removal of each edge v .

- Continue Until the Target Communities are Found: Until the network separates into discrete communities or clusters, repeat the edge removal and betweenness recalculation procedures.

Because it systematically reveals the inherent divides within a network based on the significance of edges in linking various network segments, the Girvan-Newman technique is useful for community detection.

b) **Kernighan-Lin Bisection Algorithm:** Made in 1970 by Brian Kernighan and S. Lin, it has been very widely used in many ways ever since then.

The famous graph splitting method that divides a picture into two equal parts is the Kernighan-Lin (KL) matchstick rule. To start the process, we first split the chart into two smaller groups [24]. The graph is first divided into two subgroups to begin the procedure.

The Kernighan-Lin algorithm's fundamental steps are as follows:

- Initial Partition: Dividing graphs into two sub graphs, either arbitrarily or by applying heuristics.
- Determine Initial Gain: Determine the gain in the objective function for each node in one subset if it is switched out for a node in the other subset.
- Node Pair Selection: Choose the two nodes with the highest gain, one from each subset. There will be consideration for swapping these nodes.
- Swap Nodes: In between the two subsets, swap the chosen nodes.
- Update Gains: Recalculate the impacted nodes' gains. Just the gains for the nodes that require updating due to swapping are affected.
- Repeat: Carry out steps 3-5 until either a halting criterion is satisfied or no more progress can be made
- Final Partition: Upon algorithmic termination, the final partition is obtained. A minimized objective function in the resultant partition should show a reasonably balanced division of the graph.

c) **Louvain Algorithm:**

1) **First Configuration:**

- a) Assign every node to a distinct community.
- b) Determine the network's modularity, which gauges how well the present community assignment is going .
- c) Continue until there is no more room for modularity growth (Modularity Maximization):
 - i) Take into consideration relocating each network node to a nearby locality.
 - ii) Determine how modularity might alter as a result of this relocation.
 - iii) Make the change if it improves modularity. If not, maintain the person in their present community.

2) **Create the following graph using super nodes:**

- a) In a new network, merge smaller communities into a single node. The edges of this new network

indicate all of the connections made between the communities in the previous step.

- b) On the new network, repeat steps 1.3 and 2 until modularity no longer rises noticeably.

d) **Spectral Algorithm:** Graph segmentation is the root problem of spectral clustering. Finding a partition for a graph with the least amount of edges between two disjoint groups [26] of nodes is known as the cut.

In network analysis for Alzheimer's disease, one technique employed is the Spectral Algorithm for community detection. It uses the graph Laplacian matrix's eigenvectors to find cohesive node clusters in a network. The technique works by turning the network into a spectral space, which makes communities and underlying structures visible. This technique helps reveal patterns of connectivity, pinpoint possible subgroups, and shed light on the modular structure of the disease-related network in the context of Alzheimer's research.

IV. RESULTS AND OBSERVATIONS

The network made from Individuals with and without Alzheimer's disease can be used to learn key things about how the tree works and what might change it using an important part study. Here, using the methods shown, we look into what the results from different important measures could mean.

Being able to change how we find groups in a community is very important for studying Alzheimer's disease on a network. By recognizing effective units that embody distinct features of ailment, it assists in deciphering the systematic structure of cellular pathways. This structure helps us learn more about the details of Alzheimer's, guide custom-made medicine, center treatments on an individual basis, find clues in our bodies that show if we have this disease and give understanding into how it changes over time. Generally, breaking down information into smaller parts makes it easier to understand and gives us more detailed data when we study how things are connected with Alzheimer's disease.

The amalgamation of different centrality metrics offers an all-encompassing structure for assessing the functions of distinct nodes in the Alzheimer's disease network. The highest points in each key measure show the outcomes, which give smart paths for further study and might help find significant individuals, first alerts or dangers linked with Alzheimer's illness. The different ways we can measure how connected people are to each other, through numbers or patterns, help us understand better the complex relationships and changes that occur among those affected by Alzheimer's disease.

Table I represents the centrality measures along with their values obtained from the dataset. In the same way the table II represents the modularity values that are derived from utilized community detection techniques.

A. Coreness Centrality

The coreness centrality measure offers a unique view by checking how well a node can spread among its neighbors, depending on their importance. This method, using the weak sick healed (SIR) plan, helps us learn better about how the

fast-spreading sickness moves all over a network in a harder way. To improve the accuracy of coreness-based centrality assessments, monotonicity limitations are addressed with the introduction of the μ -power community index (μ -PCI). By specifically specifying the centrality based on a node's k-core level, the k-core centrality formula enhances our comprehension even further.

B. All-around Score Centrality

Combining degree, betweenness centrality, and k-shell index, the overall score offers a comprehensive assessment of a node's impact. This measure assists in finding nodes that may be important in information propagation across the complicated network of Alzheimer's disease patients by taking into account both local and global network features.

C. Synthesize Centrality

In social networks, the synthesis centrality measure becomes very important for identifying opinion leaders. The combination of degree, betweenness centrality, and coreness centrality in this metric works well for identifying influential nodes in the context of Alzheimer's disease, where knowledge propagation understanding is critically important.

D. C-Index centrality

C-index centrality metric provides information about a researcher's competence inside a weighted cooperation network. Its relevance in comprehending the cooperative dynamics inside the Alzheimer's disease network is indicated by its link with cooperation network competency.

E. Community Detection Based on Label Propagation

A quick and easy-to-understand step-by-step plan that gives each part a different name, and keeps changing those names based on what most of the other parts around it are called. It can only find groups that don't overlap and doesn't need any settings.

F. Girvan Newman

The Girvan-Newman group finding method is a helpful way to study networks in Alzheimer's disease. By repeating to get rid of links that have a lot of betweenness centrality, this method shows the different parts in a clear way and concentrates on finding groups. Girvan-Newman's tool is handy when studying Alzheimer's because it helps find important parts, points out possible signs of the disease and shows how these body systems are put together in a way connected to this illness.

G. Kernighan-Lin

A method to split a network into two groups is by using the Kernighan-Lin method of cutting. By changing pairs of parts over and over again, it reduces the level of cut on the sides shared by both sets. We choose the sets by using a changed form of Kernighan-Lin, which evens out the splitting evenly by moving parts one at a time and switching back between sides. You can understand the complex connections and actions between different parts by using network analysis.

TABLE I
TOP NODES OF DATA SET WITH CENTRALITY MEASURES

Serial Number	Nodes	Core Number	Nodes	All-around Score	Nodes	Synthesize centrality	Nodes	C-Index Centrality
1	001	234	263	5.000026	139	0.0794	326	15.0246
2	002	234	033	5.000025	155	0.0688	374	15.0246
3	003	234	042	5.000025	449	0.0675	210	13.5259
4	004	0	244	5.000025	039	0.0578	369	13.5259
5	005	0	060	5.000024	268	0.0554	084	13.0905
6	006	0	234	5.000020	210	0.0545	449	12.8481
7	007	0	180	5.000019	369	0.0545	164	11.6775
8	008	0	205	5.000019	015	0.0520	138	11.5917
9	009	234	010	5.000013	238	0.0510	039	11.4293
10	010	234	220	5.000013	326	0.0508	212	11.3523

TABLE II
COMMUNITY DETECTION TECHNIQUES WITH RESPECT TO THEIR MODULARITY VALUES

S.No	Community Detection Method	Modularity values
1	Label Propagation	0.9672641404846366
2	Girvan-Newman	0.8011786252053879
3	Kernighan-Lin	0.4943711551773874

H. Louvian Algorithm

The Louvain Algorithm is used to study brain networks that are changed by Alzheimer's disease. Many groups are tied to different parts of AD sickness, like thinking skills, talking and remembering. Also, changes in how a group of brain cells work together have been seen more often in people with AD than those who are healthy.

This suggests that problems linking different parts of the brain could be linked to developing the disease. This method successfully spots groups of points linked very closely together, showing potential smaller patterns within the web tied to Alzheimer's disease. It helps us find nature's own parts, understand how things are put together in their simplest form and notice groups that matter for our health. These can give us clues about what leads to sickness or kinds of diseases.

V. CONCLUSION

To understand the complex workings of Alzheimer's disease, this research looked into using tools for studying networks. It studied finding groups and key points in these networks specifically. By picturing Alzheimer's disease as a group of people and their links, we could see styles and shapes that may have stayed hidden in other normal ways to find data. Looking at Alzheimer's disease groups showed us different

sets of people. These groups were made up by folks who shared some common things and they linked together. This helped in understanding the mental as well as social factors that can cause or affect how this illness starts and runs its course over time. Methods that use labels showed potential in handling groups of people or things with parts shared between them, while algorithms based on quality often found neatly separated communities.

Coreness centrality is a way to show how deep in the network's heart each node is. This points out key people important for talking and sharing information. These people could be targets of future study or help attempts. Mixing different ways to show how important a point is, the system gives us a better idea of who does what in the network.

The findings from the study suggest that we could use network analysis to learn more about Alzheimer's disease. Methods for finding groups in a community can help spot smaller parts with close disease patterns. This lets us give more specific care and direct steps to deal with these issues. Central measures aid in finding important people who are key to disease growth or the effectiveness of treatment.

VI. LIMITATIONS FUTURE WORK

In this network analysis, we focused on the available data about the participants without considering individual immu-

nity, genetic relationship between the participants, dynamic nature of the aging of disease in the study of Alzheimer's disease. Where the related data is not available in public for analysis and limited for getting out the result.

We need to study how groups of people and individual parts change with time as Alzheimer's gets worse by using a careful review method. Moreover, adding life-related information like gene and picture data to the network study plan can help us understand even better what causes Alzheimer's disease. In further studies, we need to incorporate this unavailable data that can lead to a greater insight of Alzheimer's disease.

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